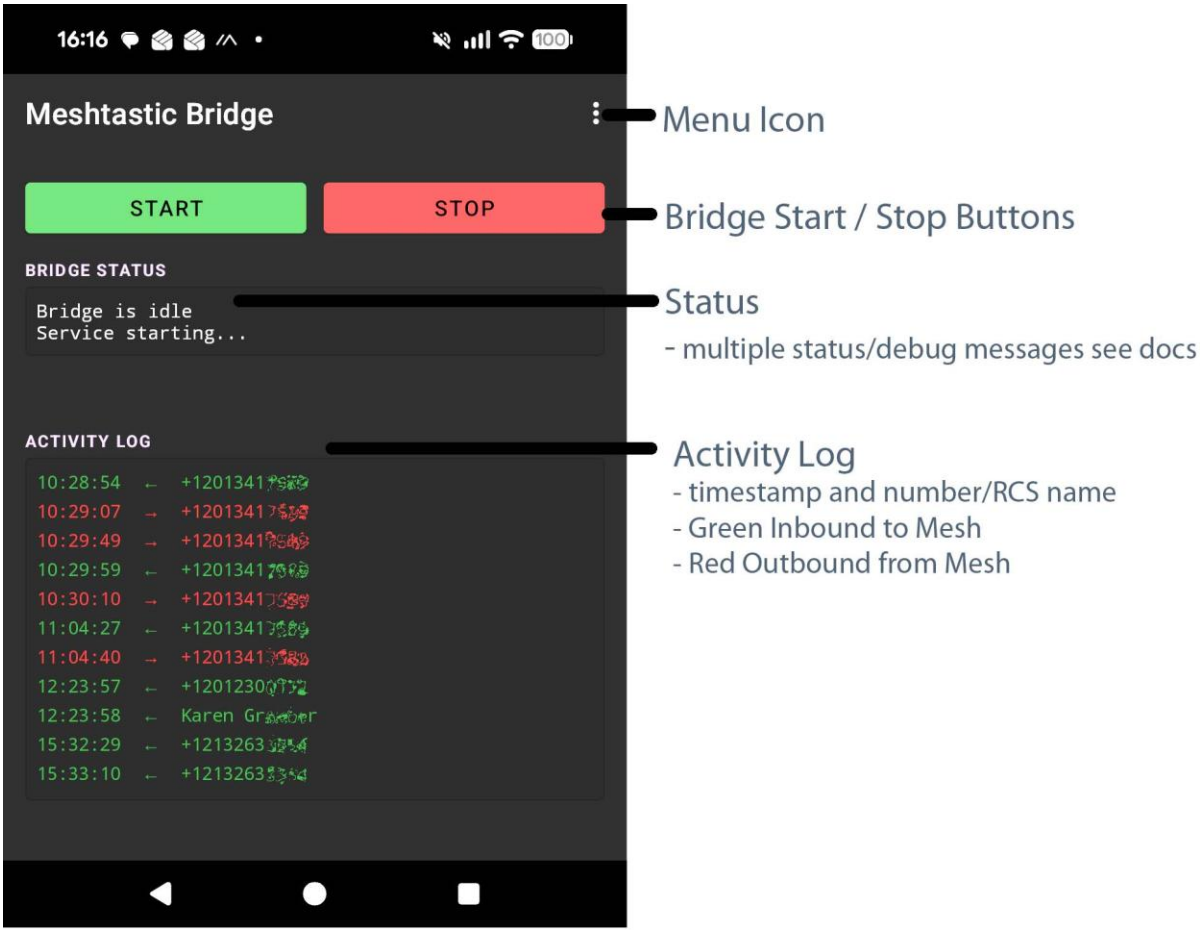
# UI Setup Guide (Alpha)

This is more a collection of notes and information in this release. This document alone should be sufficient to operate the bridge but a more comprehensive description is available on GitHub.

## 1. Main screen overview




### Main sections:

- Start: starts foreground bridge service.
- Stop: stops foreground bridge service.
- Bridge Status: rolling runtime status lines.

The available status messages are:

- Bridge is idle
- Grant Notifications permission first
- Service starting...

- Start failed - see logcat
- Service stopped
- User settings not saved: short name is required
- User settings not saved: duplicate short name '<name>'
- User settings not saved: invalid phone for '<name>'
- User settings saved: <n> caller(s), mode=<shared|channel-bound>, shared Ch=<1-10> optionally plus (preferred callers only)
- Simulated inbound SMS injected
- No mapping yet - use Debug menu to enter a phone for manual test
- Simulated mesh reply queued
- Reply queue failed

4. Activity Log: recent inbound/outbound delivery events.

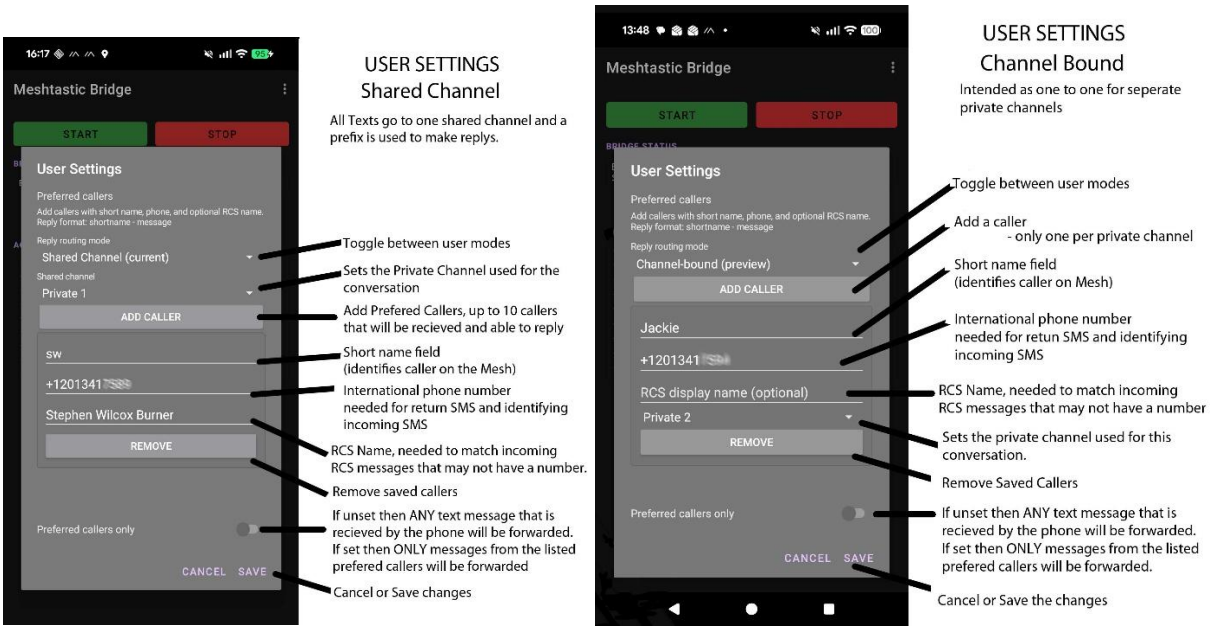
- Inbound colored Green from SMS - timestamp and phone number / RCS Caller name
- Outbound colored Red from MESH - timestamp and phone number / RCS Caller name

5. Top-right menu:

- User Settings
- System Settings
- Debug

## 2.0 User Setting

The menu option User Settings provides a way of entering and configuring Private Channels and a list of Preferred Callers plus some options that filter incoming text messages.




## 2.1 Overview and Terms

In simple terms, Private Chanels relates to the Meshtastic firmware and is used to set up how up to 7 private channels receive and send SMS/RCS traffic. It is assumed that these channels are set up as Private using a defined key shared with the Bridge Meshtastic node. A suggested convention is to use Private 1 for the shared channel and the remaining channels for Chanel Bound settings. Then configure one or multiple Bound Channels for specific callers. We plan to use Private 7 for an emergency / group chat in future releases.

- **Known issue — channel-bound mode (v0.2.0-alpha.1):**
  The current UI allows 10 private channels, only 1-7 are supported in Meshtastic. This will be corrected in the next release.

There are two types of user interface, one for the Shared Channel and one for one or more Channel Bound. Channel Bound is the more “experimental” at this stage.

The Shared Channel lets all the phone text messages flow into a single Meshtastic channel (Channel 1 is recommended) Inbound SMS traffic can be limited to only listed callers, known as Preferred callers.  This is determined by the ‘Preferred Callers Only” toggle.

You can only reply to callers in the Preferred Caller List and the number of Preferred Callers is limited to a maximum of 10.

**More clearly in the current build:**

1. Max 10 callers assigned to the shared traffic pool.
2. Dedicated private channels are single-caller only.
3. A dedicated caller cannot also consume a slot in the shared pool.
4. No duplicate callers across categories.

**Guidance:**

1. Use Shared Channel for simplest all or a defined list of callers text messages sent to a single Meshtastic Private Channel and able to reply to those listed as Preferred Callers..

2. Use Channel-bound when you want different callers routed through to different private channels. This allows you to reply without using a prefix and have separate Private Channels for each Preferred Caller.

With only one caller, Shared mode can look similar, but it is not fully the same as Channel-bound mode.

1. SMS to mesh channel selection differs.
   In Shared mode, outbound always uses the shared channel setting
   In Channel-bound mode, outbound uses the caller's assigned channel if present, otherwise falls back to shared channel
2. Mesh to SMS handling differs on channel 0.
   Channel-bound explicitly rejects public channel inbound replies
   Shared mode does not do that rejection branch.
3. Mesh to SMS target resolution differs.
   Channel-bound tries latest phone mapping for that specific inbound channel first
   Shared mode ignores channel context and, for plain text replies, uses the single preferred caller globally
4. Configuration model differs.
   Caller lists are stored separately per mode, so switching modes changes which caller table is active.

- **Known issue — channel-bound mode (v0.2.0-alpha.1):**
  If you assign two or more callers to the same Private channel in channel-bound mode, inbound mesh replies for that channel will be silently dropped. Work-around: each caller must be assigned to a different Private channel. This will be validated and blocked in the next release.

- **Current limitation: only one routing mode is active at runtime.**
    - Behavior detail: last saved mode is active.
    - Workaround: treat modes as presets and verify active mode before operations.
    - Tester feedback request: gather which hybrid precedence model is most useful in the field.
    - Planned direction: evaluate a true hybrid mode with explicit precedence rules.

**The Preferred Caller list has three main entries,**

1. a short name used to identify the messages in the Mesh environment and as a way of prefixing replies from the Mesh in the Shared Cannel mode.
2. A full phone number that is needed for all replies from the Mesh because RCS does not provide a phone number which is needed to make SMS replies.
3. The RCS text identifier which is needed to match some types of messages. This is the name you gave your incoming caller in your phone directory.

Some background on the Preferred Call List requirements:

**Plain-language reason:**

1. SMS often has no friendly sender name, and RCS names can be long or inconsistent.

2. The bridge needs one stable, easy-to-type identity to route replies reliably.

3. Short name is that stable identity. It becomes the operator-facing handle for that caller.

How identity is normalized:

1. Inbound sender can be recognized from phone number or optional RCS display name.

2. Once matched, the app maps that sender to the row short name.

3. Replies use the short name route, not the raw RCS label or missing SMS name.

Example:

1. RCS inbound from `Johnathan M. Rodriguez (Work)` maps to short name `jr`.

2. SMS inbound from `+12013417589` (no display name) also maps to `jr` if phone matches.

3. Operator replies as `jr - copy, meet at checkpoint` where jr – is the prefix used to clarify who to reply to.

**Preferred callers table**

For each row:

1. Short name: compact operator tag (example: `sw`).

2. Phone: full international format (example: `+12013417589`).

3. RCS display name (optional): contact display name from notification.

4. Channel (channel-bound mode only): Default / Private 1..10.

5. Remove button: deletes this row.

**Add Caller**

1. Add Caller: adds a blank row.

2. Save: validates and saves current mode list.

Validation rules:

1. Short name required.

2. Short names must be unique in current mode list.

3. Phone must normalize to 7+ digits (with optional leading +).

**Preferred callers only toggle.**

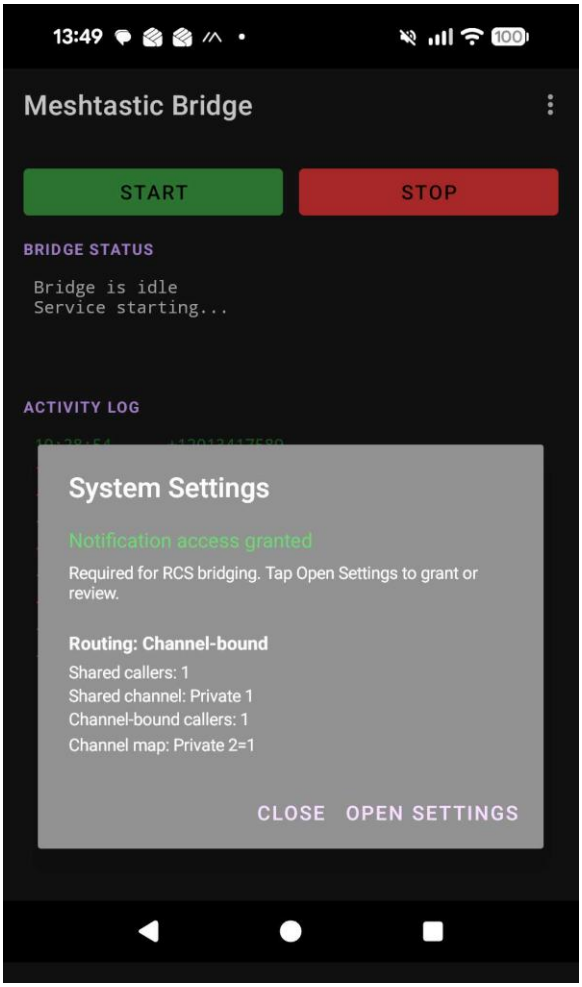
Behavior:

1. ON: Inbound Text Messages are filtered to allow ONLY those listed in the Preferred call list, so inbound SMS/RCS must resolve to a preferred caller.

2. OFF: All inbound text messages from the phone are passed to the Mesh Private Channel, so inbound is allowed even if caller is not in preferred list.

## 3.0 System Settings

The system menu summarizes the status and configuration of the bridge. It also facilitates setting the permissions and phone system settings.



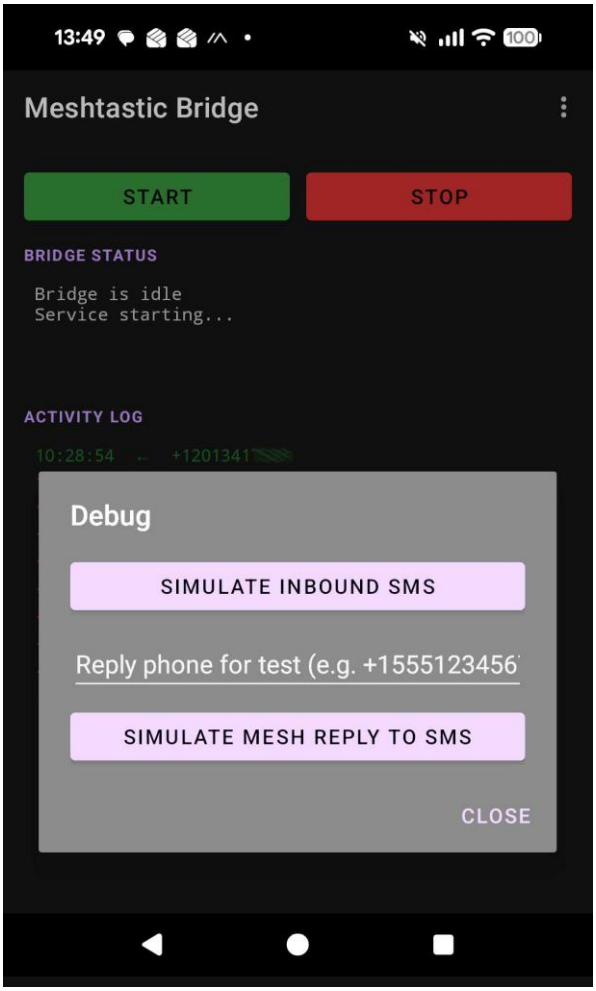

Current information shown:

1. Notification access status.
2. Active routing mode.
3. Shared caller count.
4. Shared channel.
5. Channel-bound caller count.
6. Channel map summary.

The Open Settings button opens Android Notification Listener settings.

## 4.0 Debug Menu

This menu has tools for helping debug the system.




There are two options for either sending or simulating a received SMS. This is helpful when first setting up the bridge.

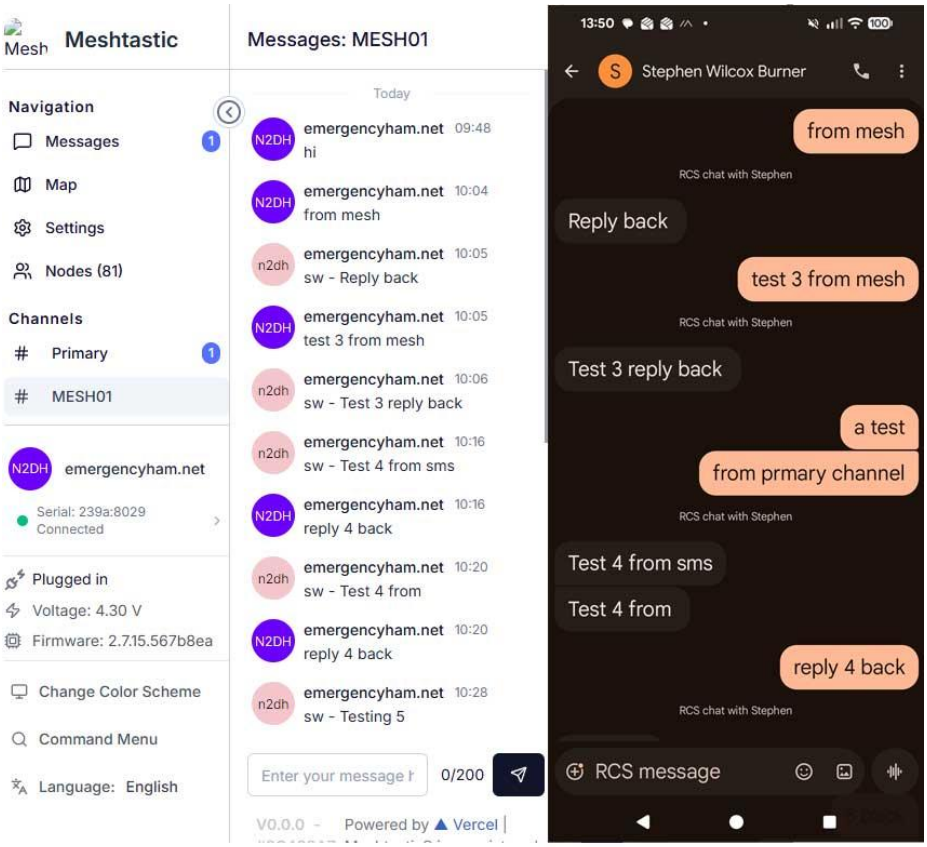
Actions:

1. Simulate Inbound SMS.

2. Simulate Mesh Reply to SMS (optional target phone input).

## 5.0 Example Use Case

This is an example of a shared channel set up with just one preferred caller, the trick here is that since there is only one preferred caller the normally expected reply prefix is not needed as there is no ambiguity to reply to. Ther is an example of a normal RCS message also being transferred, although this message cannot be replied to.




MESH Phone